

NeuFlow v3

Optical flow you can query,
with a confidence value attached

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The three objectives set at the start

- **Better accuracy** than NeuFlow v2
- **Less compute** than NeuFlow v2
- **Runs on edge devices**

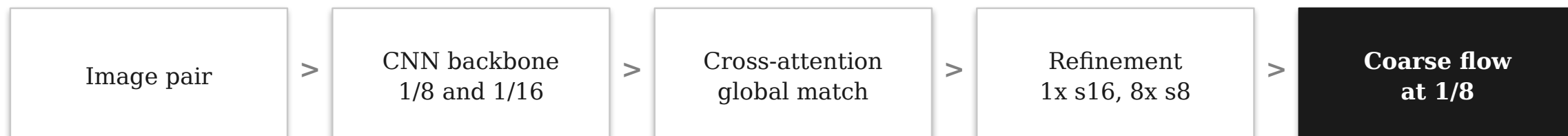
A fast platform cannot afford the frame it is given



- Registration, tracking and mosaicking each pick their own few hundred points.
- Flow between pixels cannot be requested at all.
- A second question about the same frame costs the whole computation again.

Only the last stage changes

PHASE 1 once per frame pair, cached



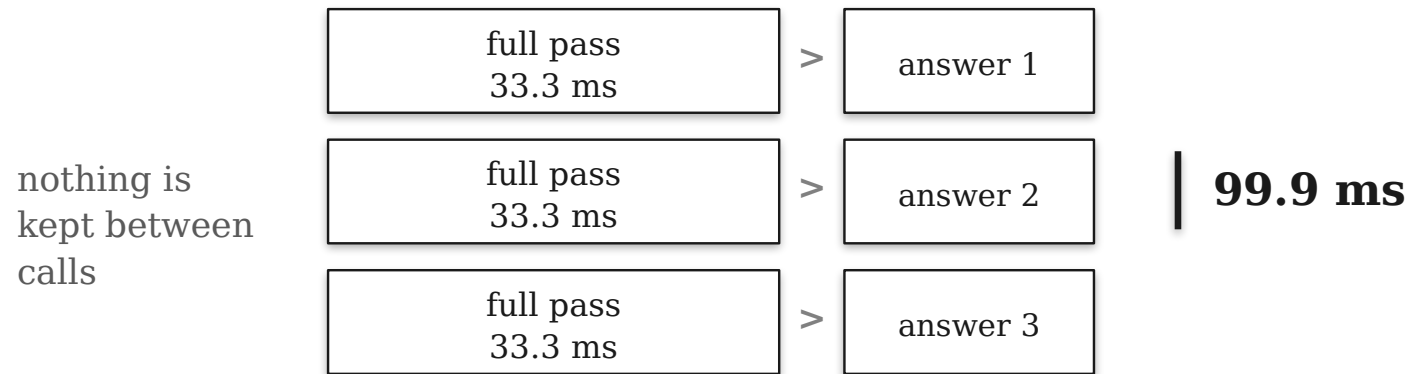
PHASE 2 once per query batch, $O(N)$



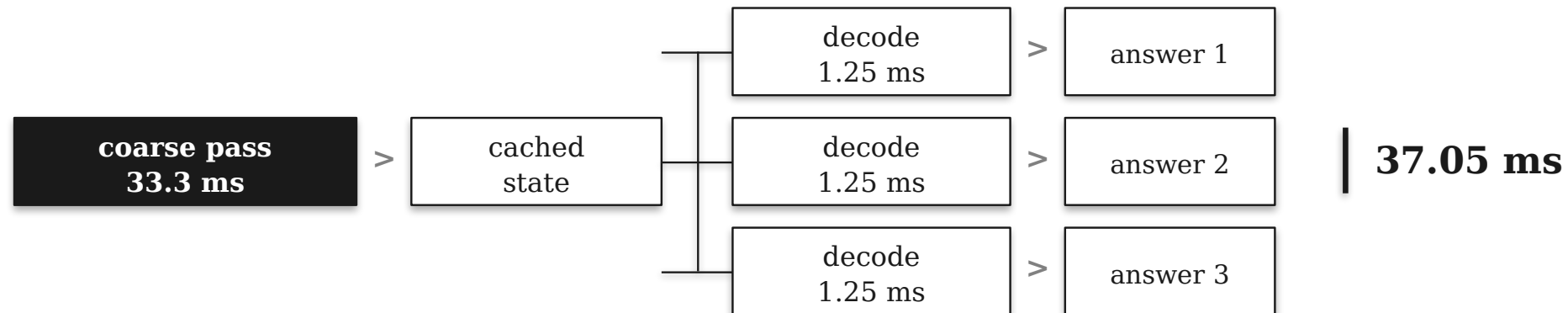
- Everything in Phase 1 is frozen and verified bit-identical to v2.
- A convex blend cannot predict motion its inputs do not support.

Three questions about one frame

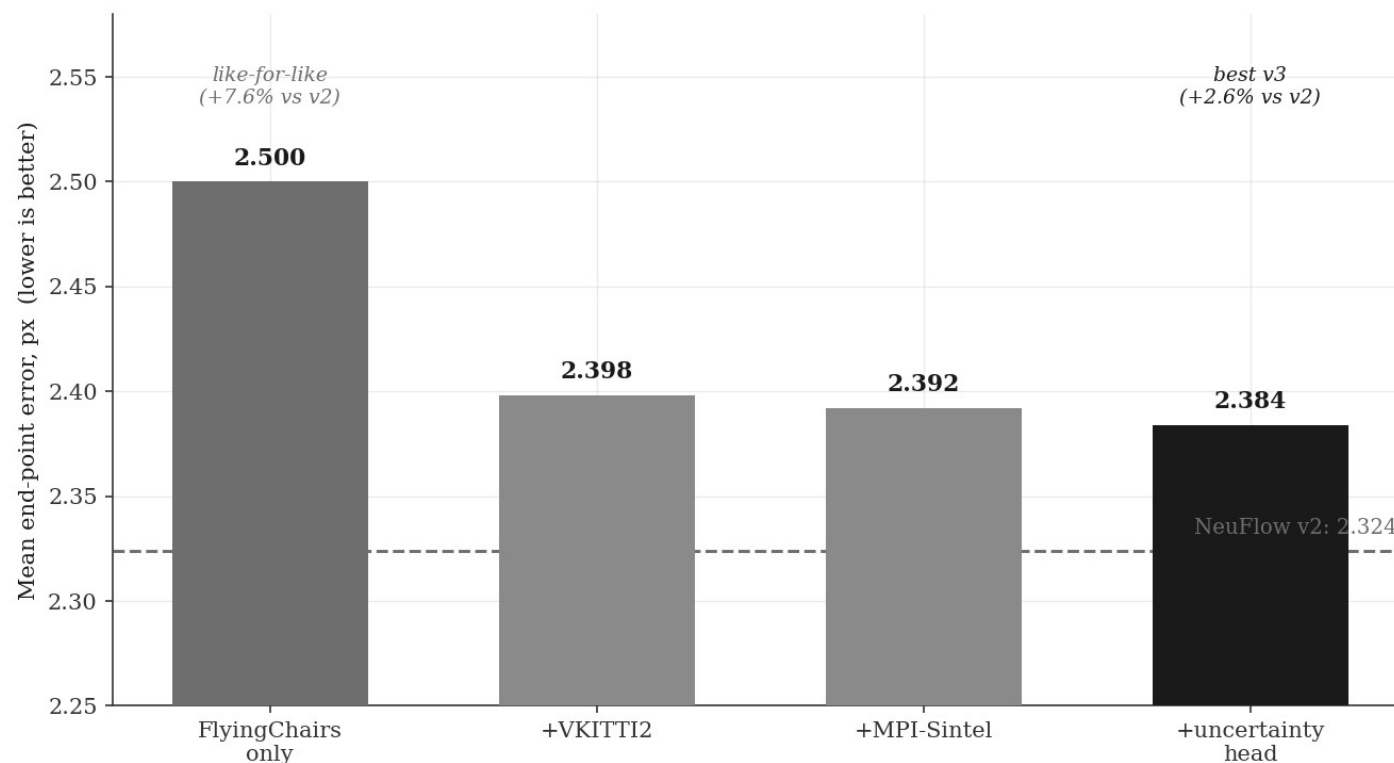
NEUFLOW V2



NEUFLOW V3



The price: 2.6% of mean error

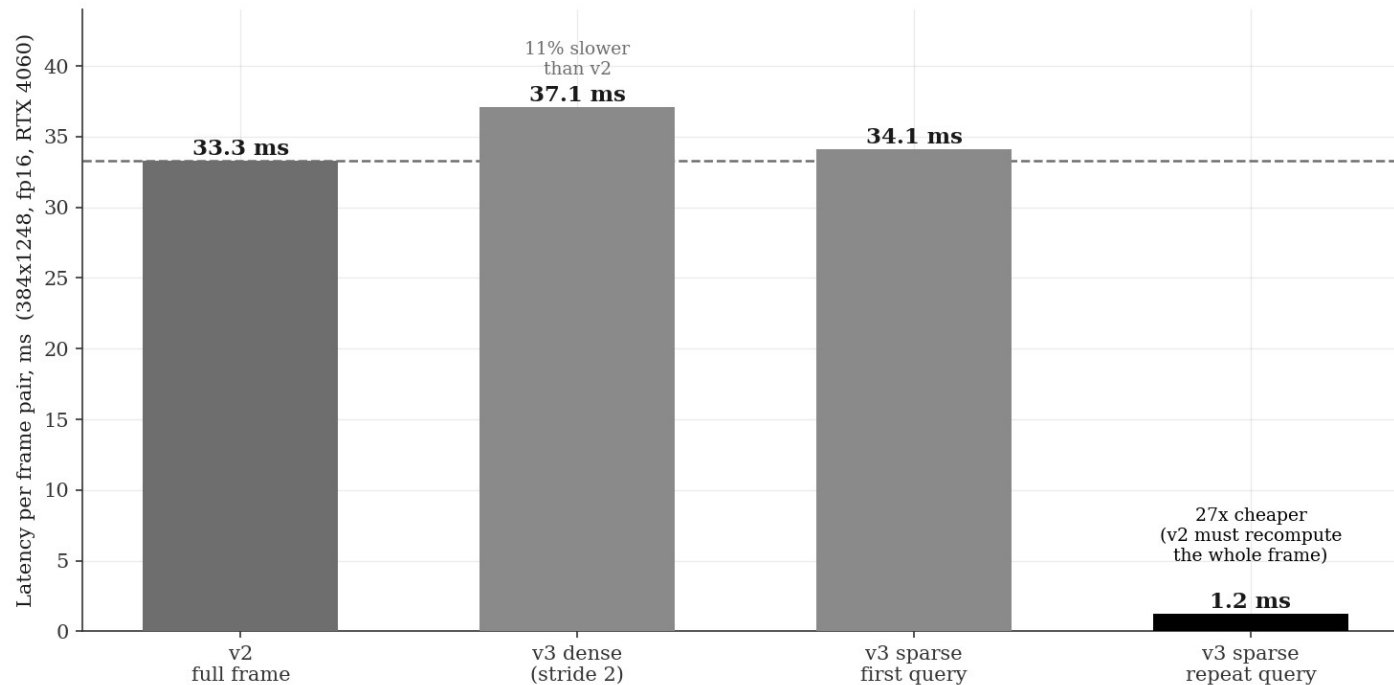


WHAT THIS SHOWS

- Every configuration sits above v2.
- Like-for-like: 2.5, or 7.6% behind.
- 7.83 M against 9.03 M parameters.

2.384
against 2.324

A repeat query costs 1.25 ms

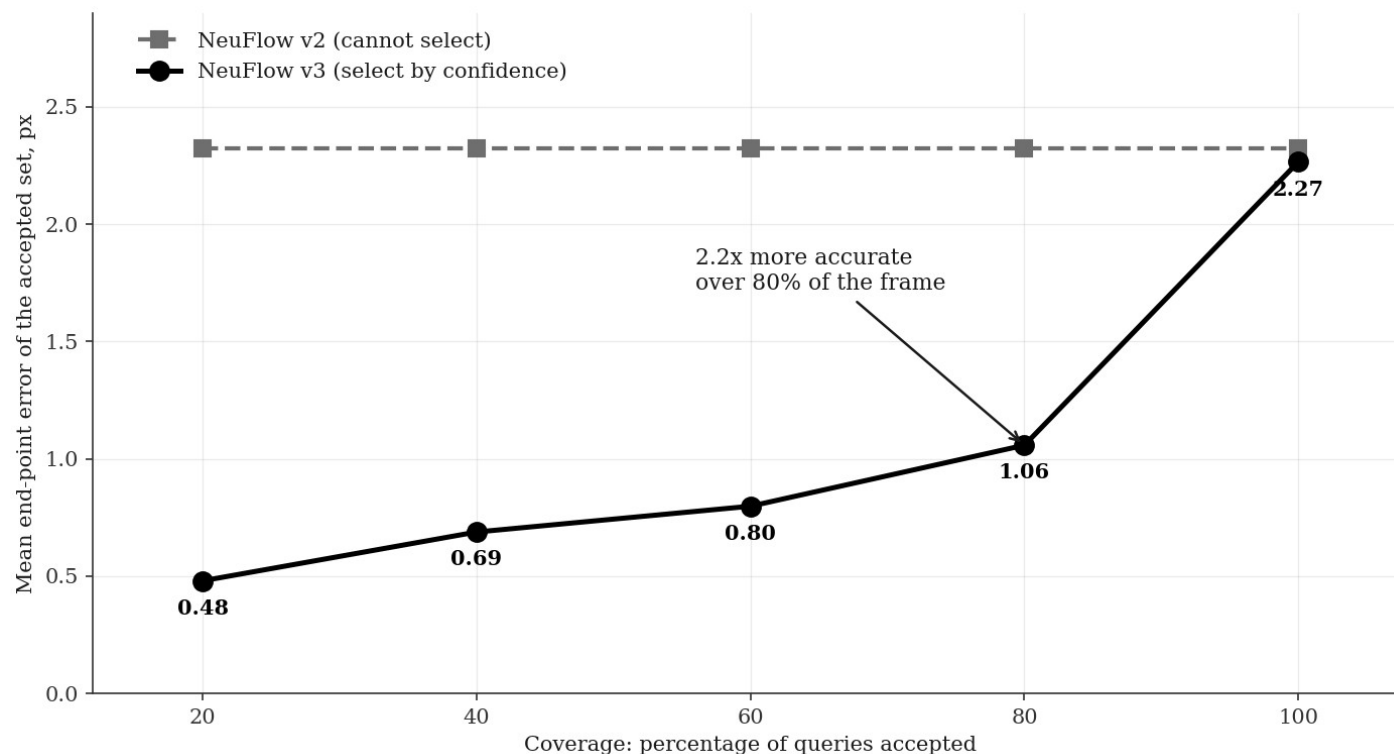


WHICH CLAIM I MAKE

- Dense output is 10% slower.
- A first query is level: 96% of it is the shared coarse pass.
- The win is the second question.

27x on repeat access

Confidence buys the accuracy back



WHAT THIS SHOWS

- Error rises monotonically across five bins, 0.48 px to 7.10 px.
- v2 has one operating point; v3 has a curve.
- Abstaining is free when the consumer picks its own points.

**1.058 px over
80% of the frame**

Flow 14% of the frame, not all of it



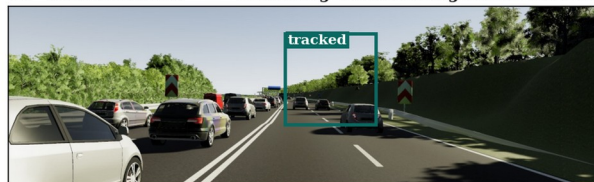
- Processing 14% of the frame costs 0.034 px and runs 4.4x faster.
- Without a margin, error rises 65% and a quarter of large-motion pixels fail.
- Design rule: margin $\approx$ expected motion $\approx$ speed / frame rate.

The same three situations, two domains

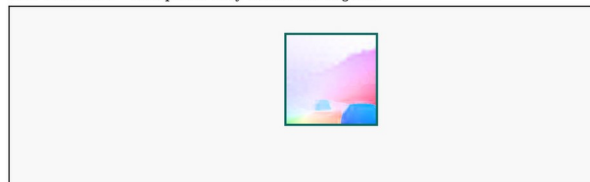
Driving

Virtual KITTI 2, 26.6 px mean motion

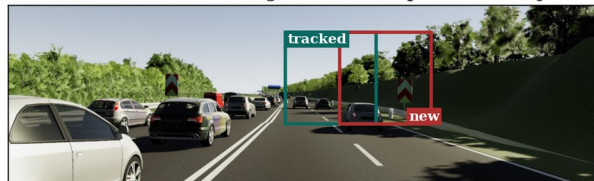
S1 First encounter — something worth flowing enters the field of view



flow returned — computed only inside the regions



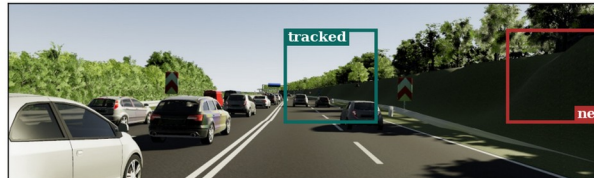
S2 Turn — the tracked region now overlaps a second object



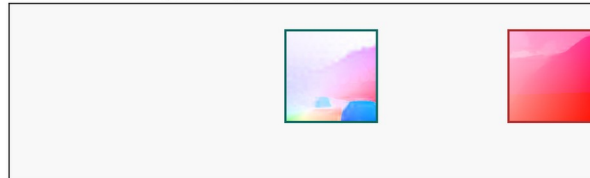
flow returned — computed only inside the regions



S3 New object — a second region appears in a frame already being processed



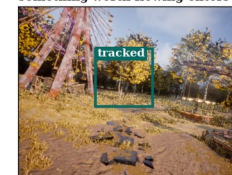
flow returned — computed only inside the regions



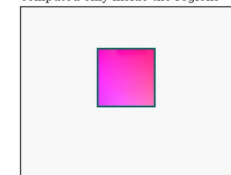
Aerial

TartanAir, 9.26 px mean motion

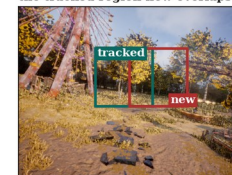
S1 First encounter
something worth flowing enters the field of view



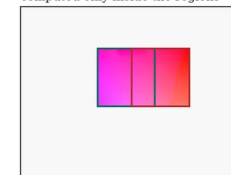
flow returned
computed only inside the regions



S2 Turn
the tracked region now overlaps a second object



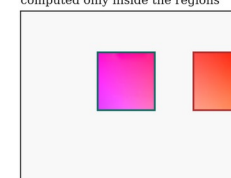
flow returned
computed only inside the regions



S3 New object
a second region appears in a frame already being processed



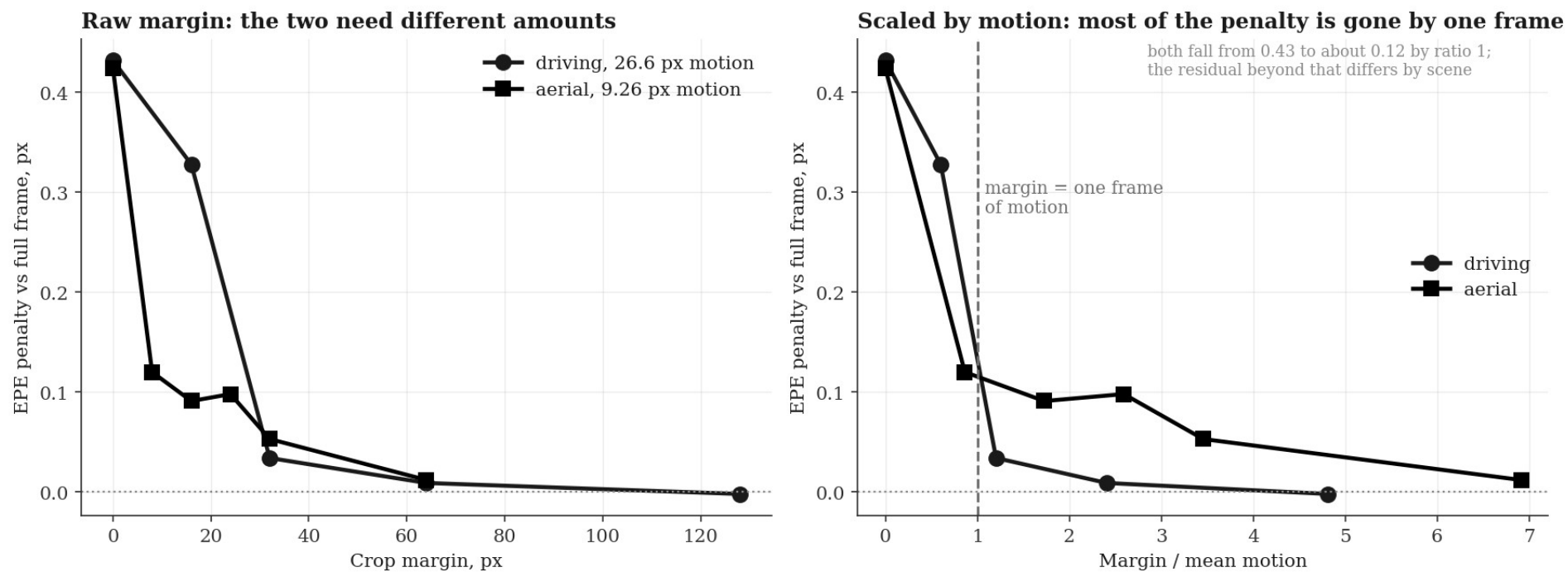
flow returned
computed only inside the regions



- Region geometry, margin and decoder configuration are identical across both.
- Only the domain changes, and no model in this study trained on aerial imagery.

Left: Virtual KITTI 2. Right: TartanAir. Flow colour encodes direction.

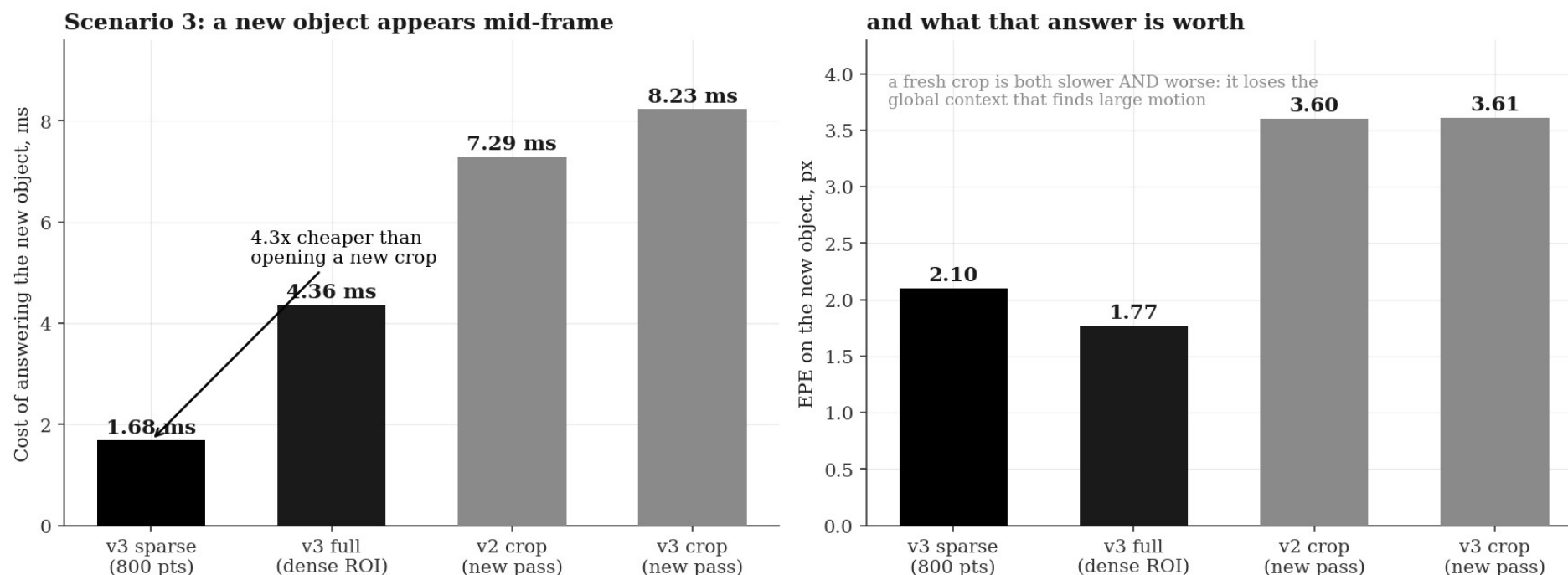
Margin is set by motion, not image size



- **Driving.** 26.6 px mean motion, needs about 32 px.
- **Aerial.** 9.26 px mean motion, needs about 8 px.

Measured with the baseline network. Predicts the scale, not the exact curve.

A region found mid-frame costs 1.7 ms



- A fresh crop runs a whole new pass at 7.3 ms, and is less accurate doing it.
- Two disjoint crops cost more than one full frame: crop once, or not at all.

One cause, three experiments converging



- Fourier positional encoding, sub-pixel querying and reduced-resolution querying all returned null, and one mechanism explains all three.
- **Remedy.** A full-resolution stem, roughly 1 to 2 ms. Not yet built.

Back to the three objectives

- **Better accuracy than v2.** Not met. 2.384 against 2.324.
- **Less compute than v2.** Partly. Dense slower, repeat query 27x cheaper.
- **Runs on edge devices.** Unproven. Every figure is from a laptop GPU.

DELIVERED INSTEAD, AND MEASURED

Queryable output

any continuous
coordinate

Cheap repeat access

1.25 ms per
further query

Calibrated confidence

a per-query
error estimate